



Team 7

Traffic Congestion Level Prediction

Using Random Forest & Gradient-Boosted Trees

Team Members:

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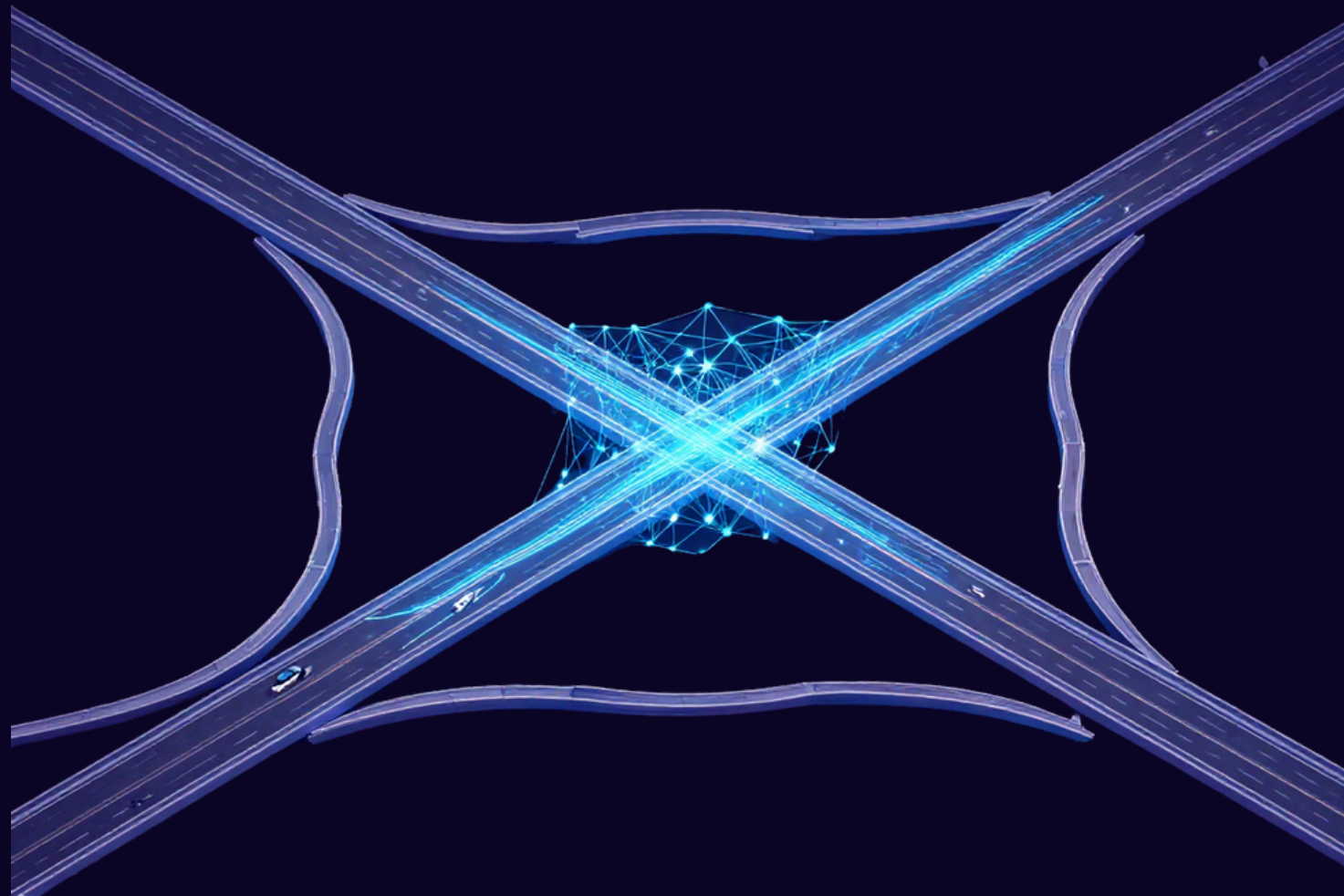
E. Nakshatra | 2520030090



Machine Learning | Section 08

Guide : Shaik Asif

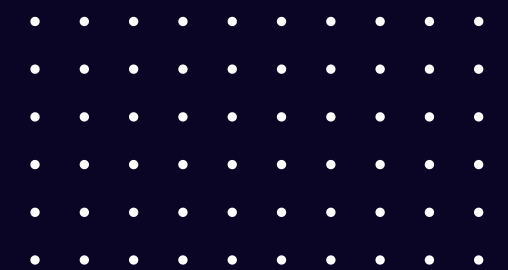
Assistant Professor CSE



Problem Statement

- Increasing vehicles and limited road infrastructure cause severe traffic congestion.
- Congestion results in:
 1. Increased travel time
 2. Fuel consumption
 3. Air pollution
 4. Economic losses
- Traditional systems mainly monitor traffic, rather than predicting future congestion.

Our Goal:
Predict traffic congestion levels using Machine Learning.



Solution

Our proposed system uses Machine Learning to predict traffic congestion levels by analyzing multiple traffic and environmental factors.

Input Factors

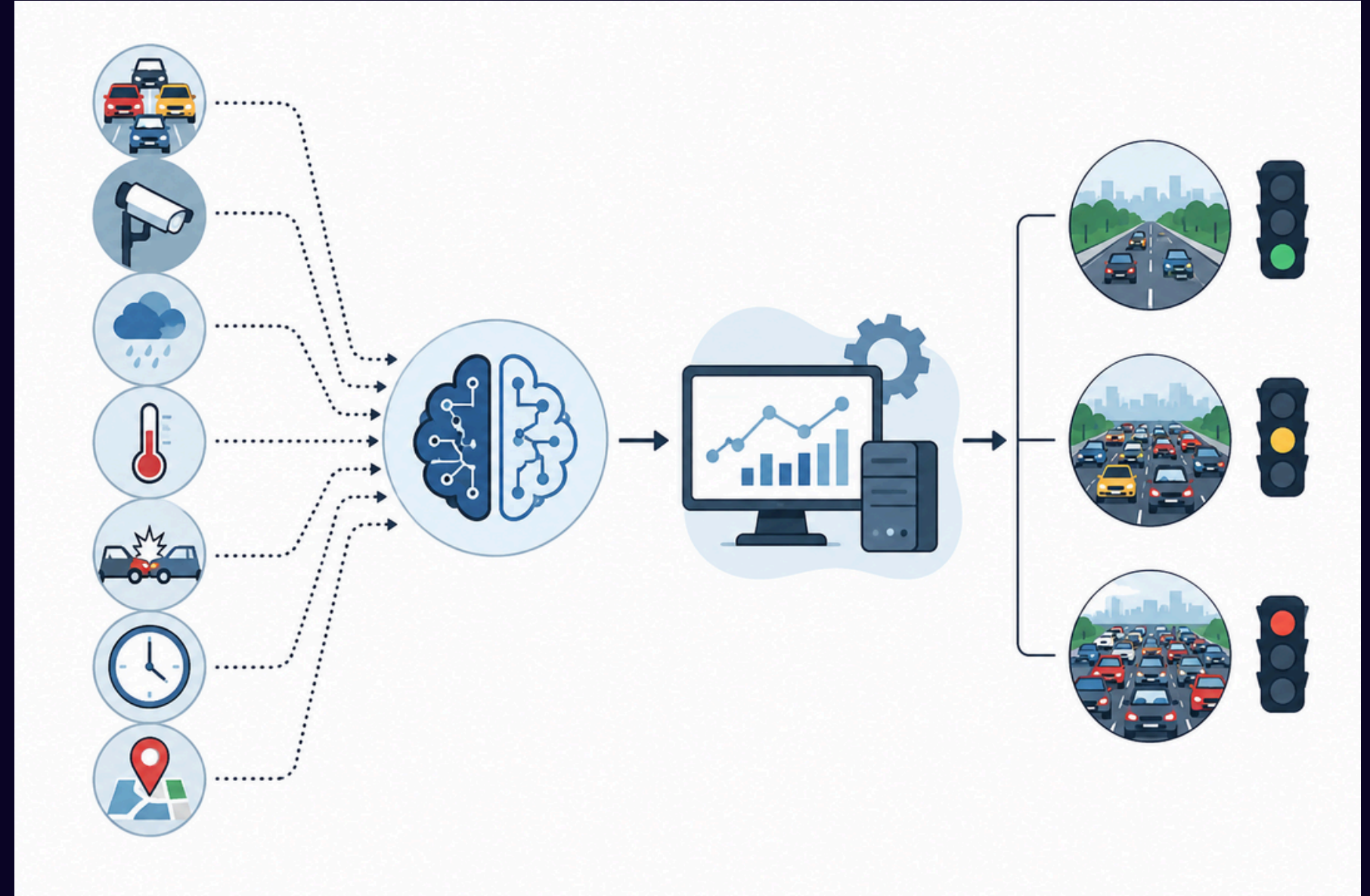
- Traffic Volume
- Average Vehicle Speed
- Vehicle Counts
- Weather Conditions
- Temperature & Humidity
- Accidents
- Time & Location
- Traffic Signal Status

Machine Learning Models

- Random Forest
- Gradient-Boosted Trees

Output

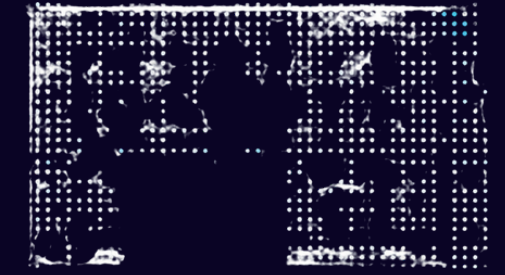
🚦 Traffic Congestion Level
Low | Medium | High



Compare both models and identify the better-performing approach for traffic congestion prediction.



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Objectives

- Predict traffic congestion levels accurately.
- Analyze important traffic and environmental factors.
- Implement Random Forest and Gradient-Boosted Trees.
- Compare the performance of both algorithms.
- Develop a system that can support better traffic planning and management.

Traffic Congestion Level Prediction



Dataset Overview

Smart Traffic Management Dataset Source: Kaggle

Features:

- timestamp
- location_id
- traffic_volume
- avg_vehicle_speed
- vehicle_count_cars
- vehicle_count_trucks
- vehicle_count_bikes
- weather_condition
- temperature
- humidity
- accident_reported
- signal_status
- congestion_score
- congestion_level ← Target
- hour
- day_of_week
- is_weekend

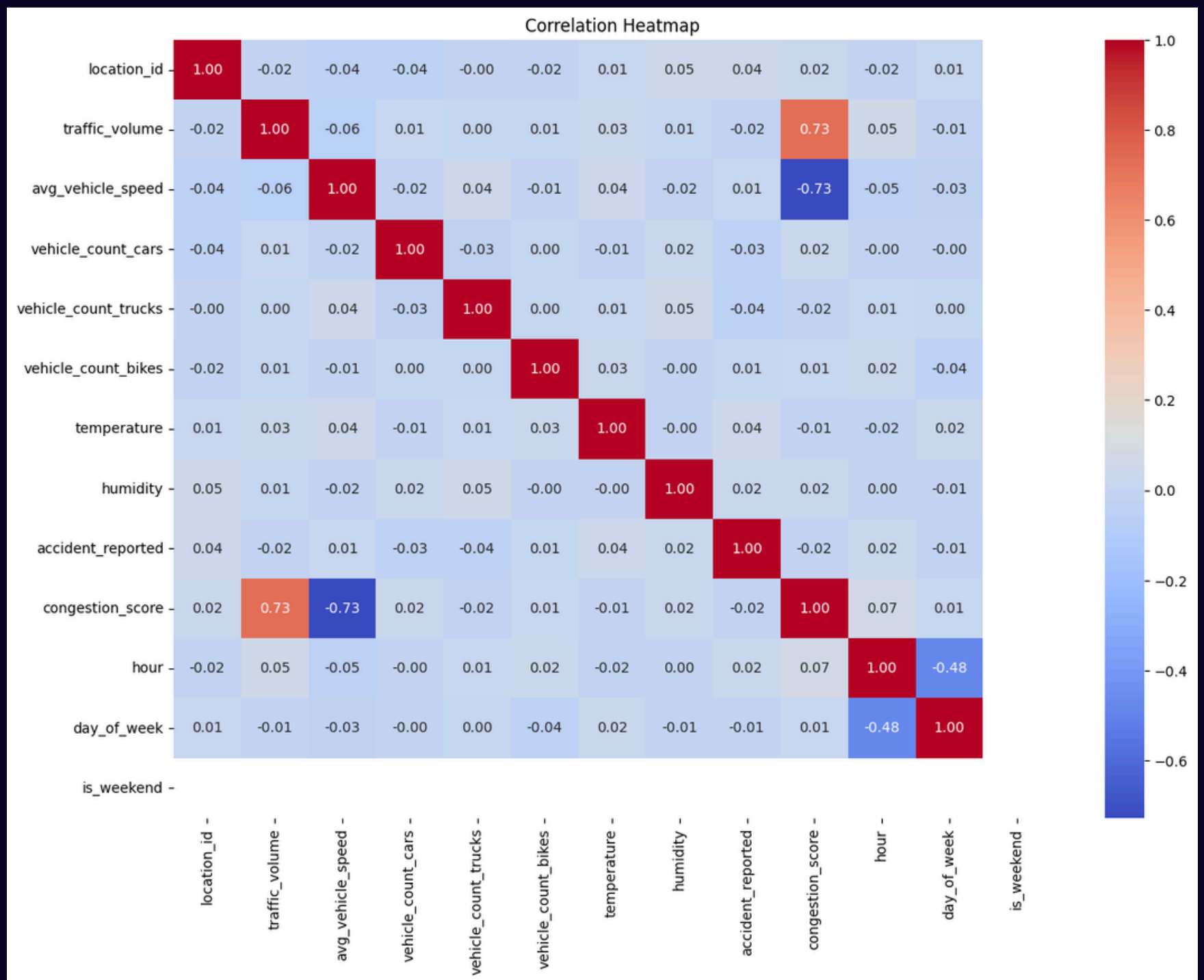
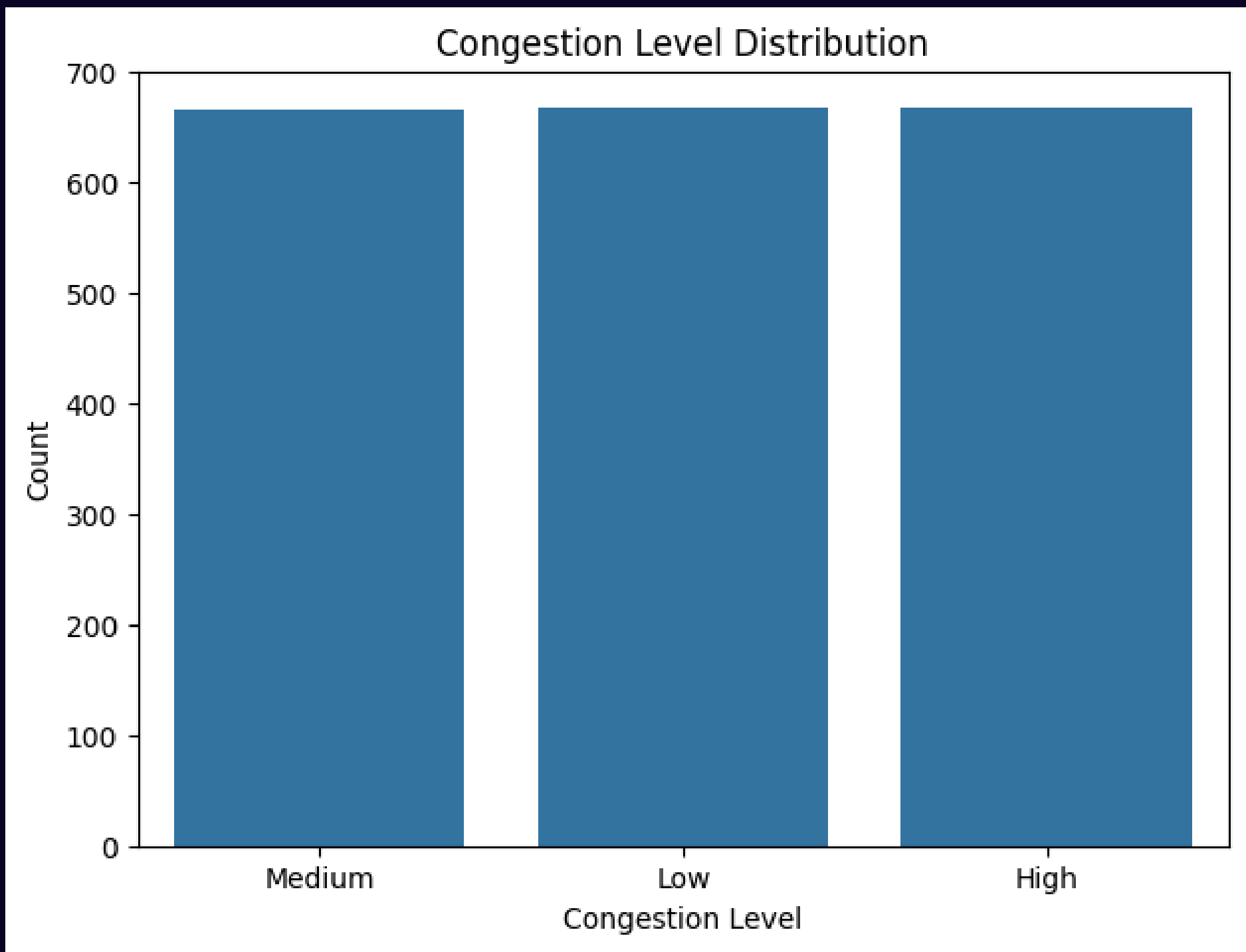
Dataset Size:

- 2,000 rows
- 17 columns



FDA was performed

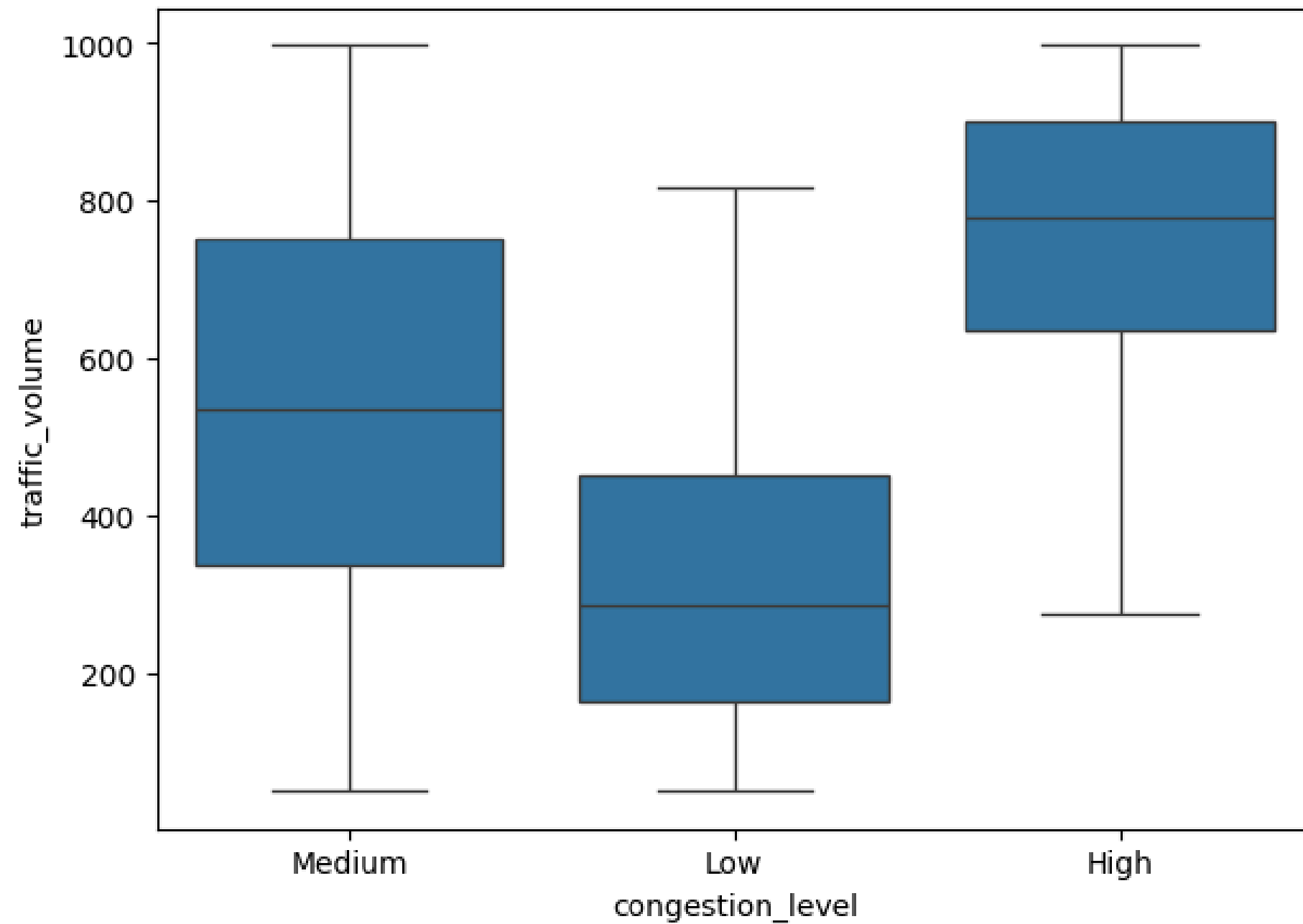
between traffic-re



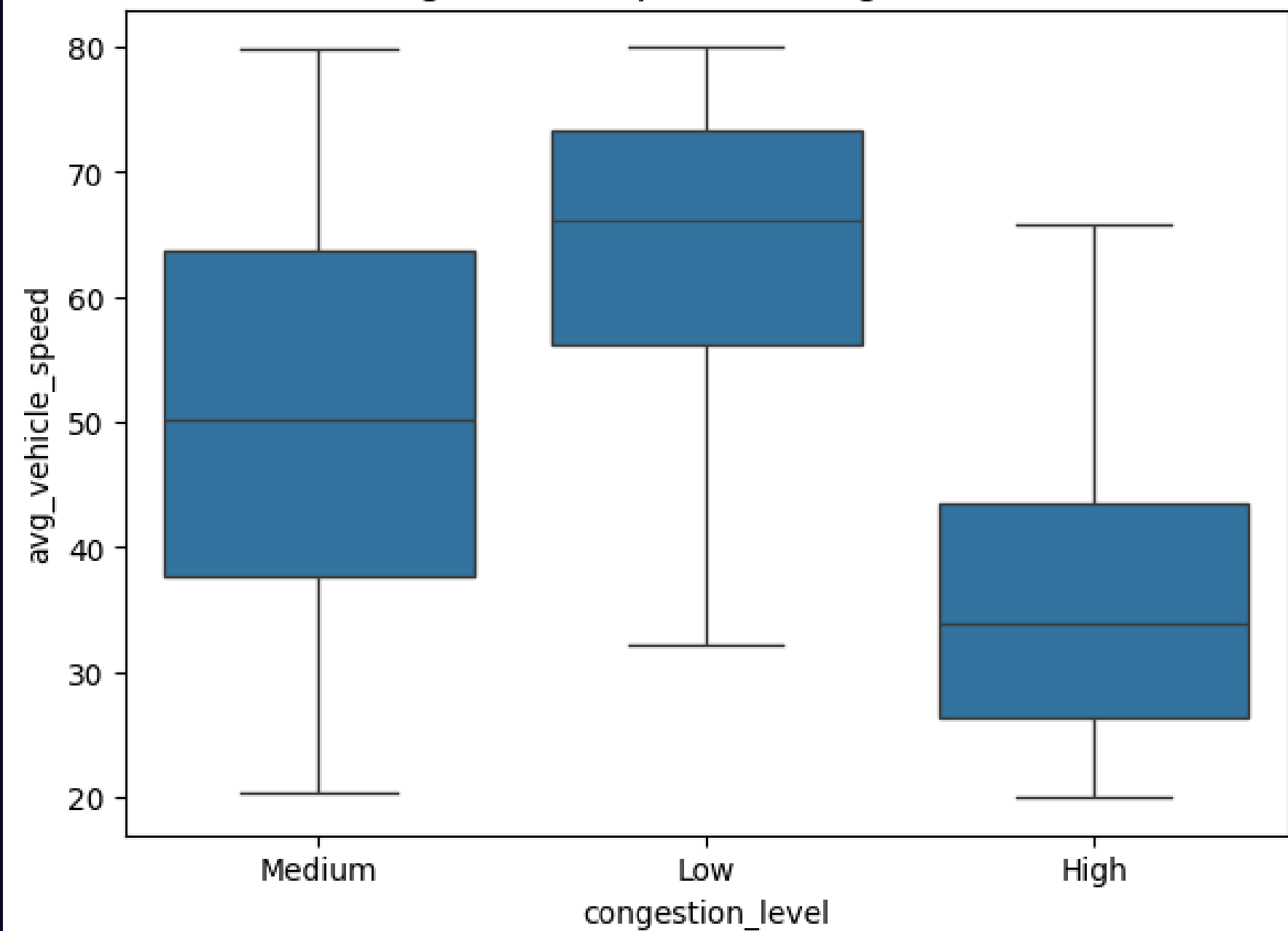
EDA Findings:

- No missing values were found in the dataset, so no imputation was required.
- No duplicate records were present.
- The target variable congestion_level is almost perfectly balanced:
 - Low – 667
 - Medium – 666
 - High – 667
- Traffic volume varies across congestion levels, showing its relevance for congestion prediction.
- Average vehicle speed shows differences across congestion levels, with congestion-related variation visible in the distributions.
- The correlation heatmap helped identify relationships among numerical traffic features.
- Boxplots and histograms were used to examine data distribution, spread, and potential outliers

Traffic Volume vs Congestion Level



Average Vehicle Speed vs Congestion Level



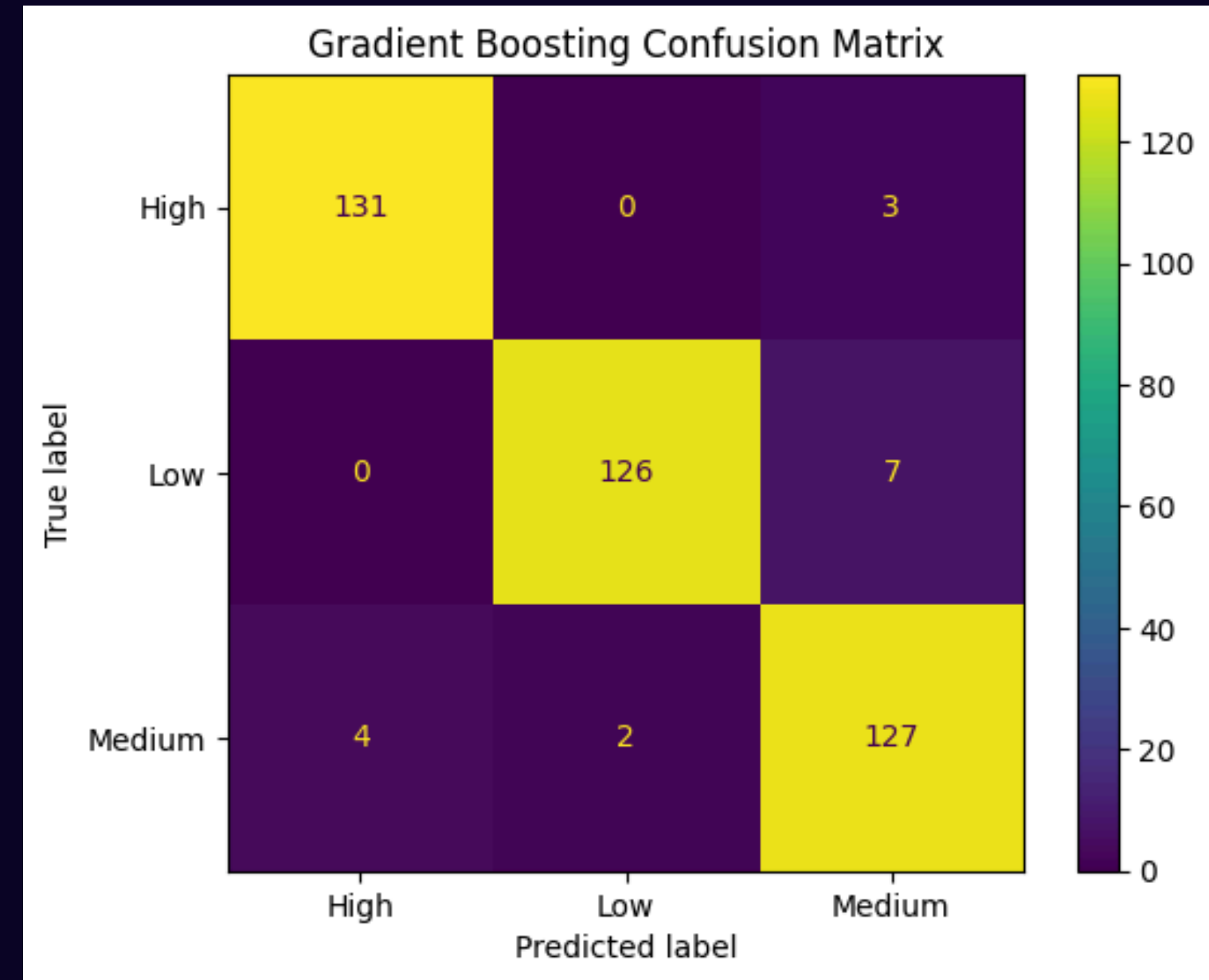
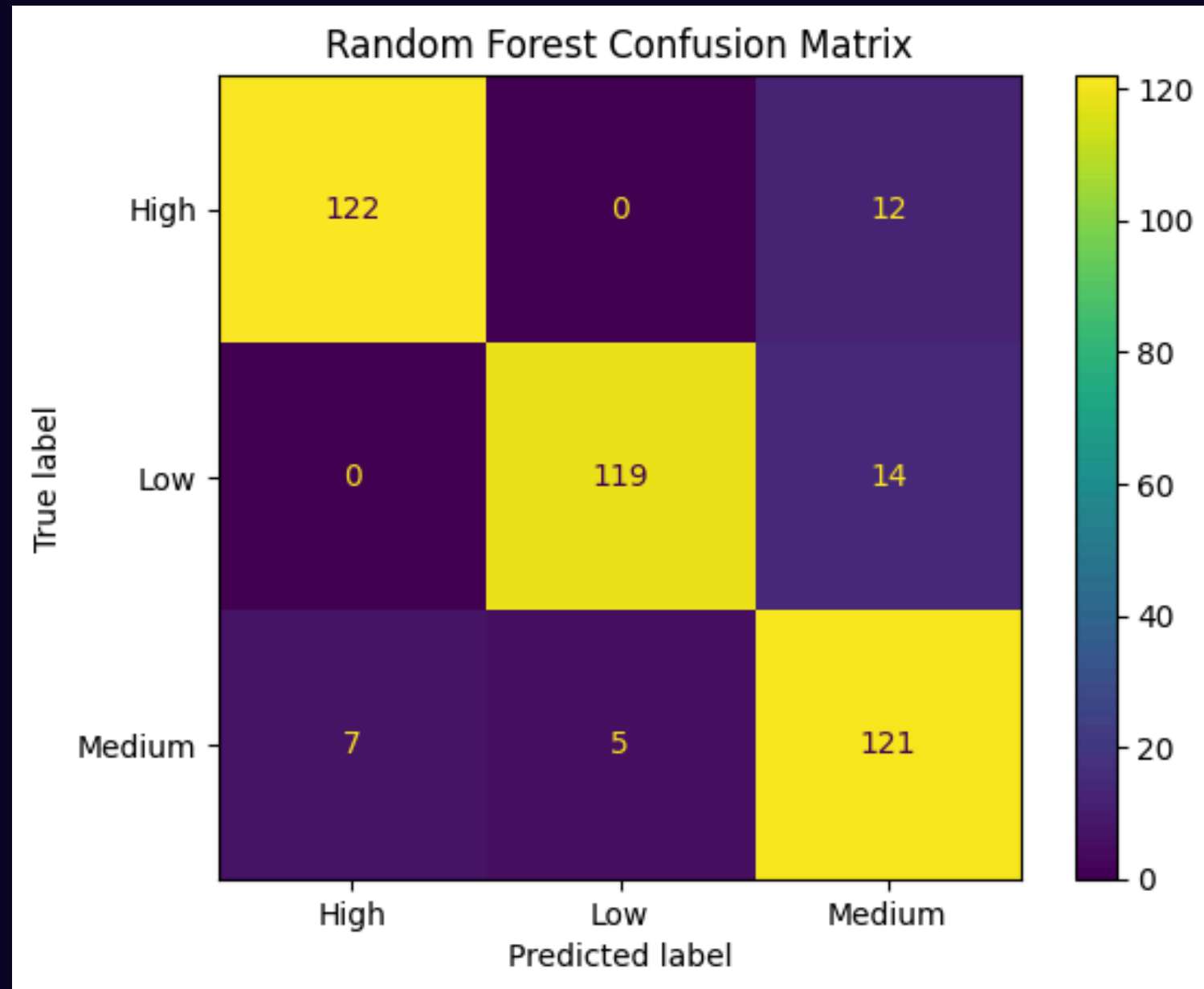
Research Papers / Literature Survey

Research Paper	Algorithm	Key Finding
Traffic Congestion Prediction Based on ETA	Random Forest	Random Forest showed effective congestion classification.
Ensemble Learning for Short-Term Traffic Prediction	Gradient Boosting	Gradient Boosting performed well for traffic prediction.

Drawbacks & How We Overcame Them

- Limited / location-specific data → Used multiple traffic, weather, vehicle and time-related features.
- Model performance limitations → Compared Random Forest and Gradient Boosting.
- Limited model optimization → Applied GridSearchCV and 5-fold cross-validation.
- Data quality issues → Performed EDA and verified no missing values or duplicates.

Results



A confusion matrix compares the actual labels with the model's predicted labels.

-  Brighter colours → more correct predictions.
-  Darker colours → fewer predictions / misclassifications.
- Diagonal values → correctly predicted classes.
- Off-diagonal values → incorrect predictions.

What it means here: Both models have most values on the diagonal, showing that they correctly classify High, Low, and Medium congestion levels, with only a small number of misclassifications.

Result

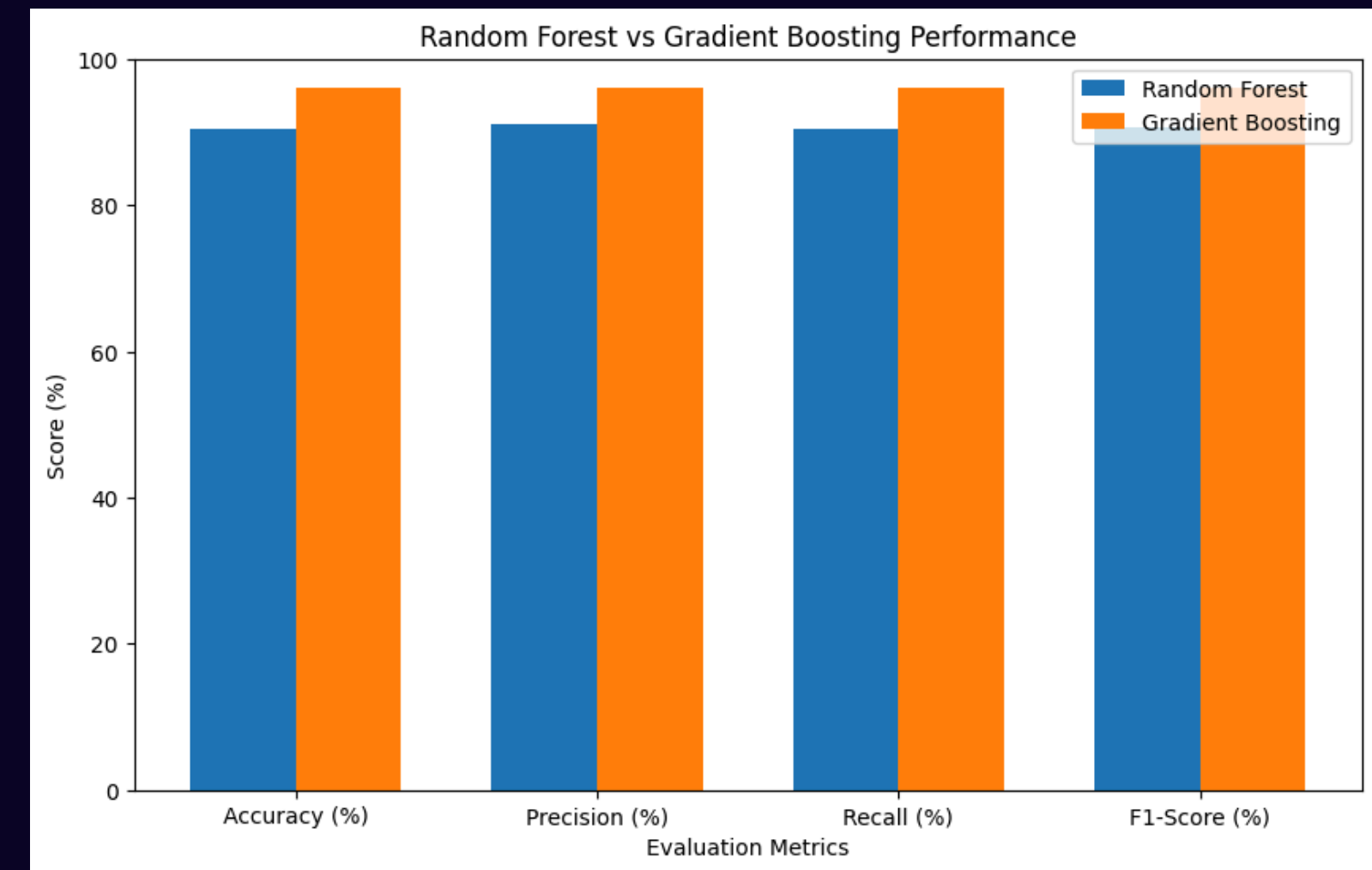
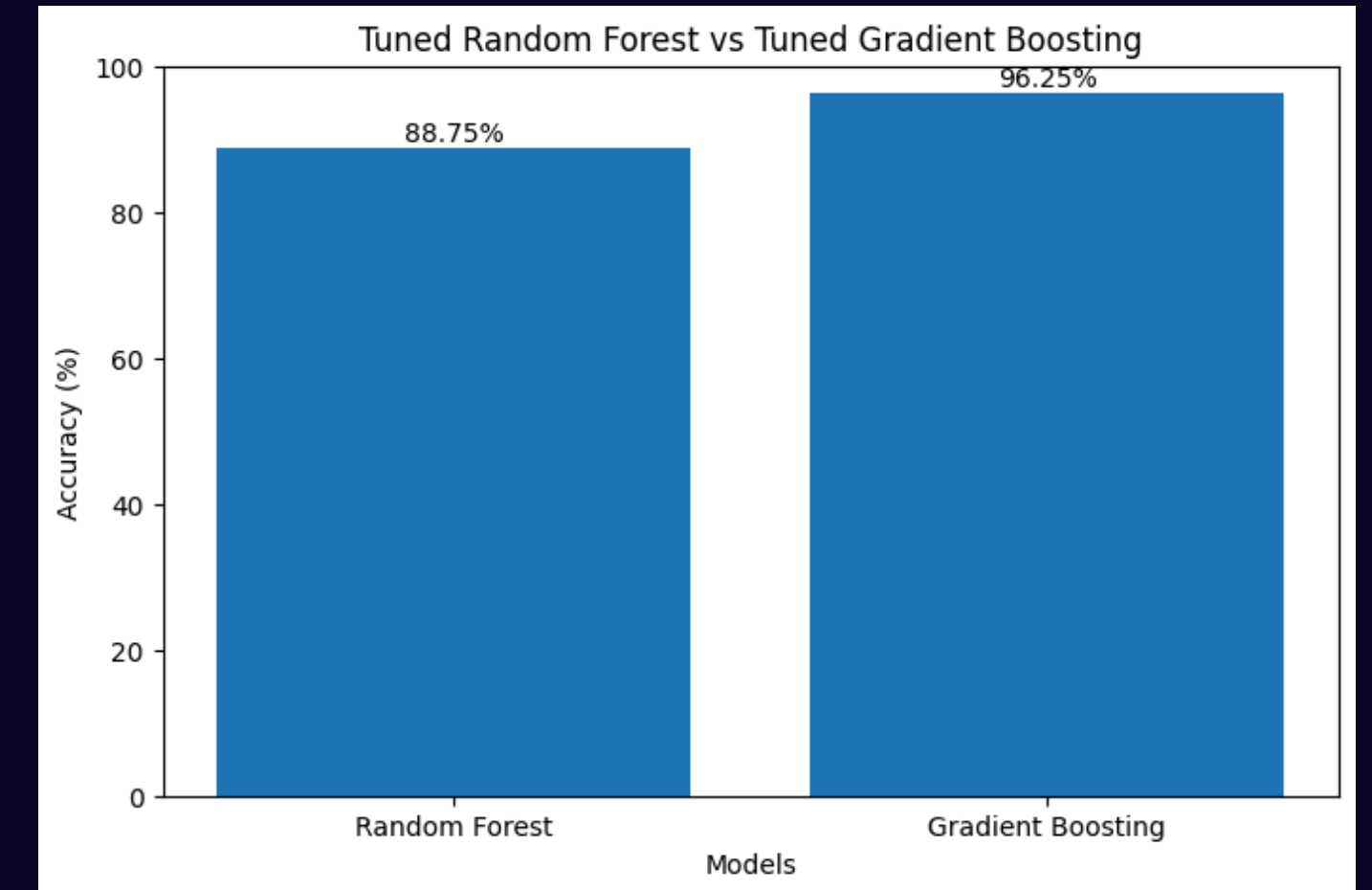
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✓ Os
▶ print("=" * 60)
  print("TRAFFIC CONGESTION PREDICTION - FINAL SUMMARY")
  print("=" * 60)

  print("Dataset: 2000 records, 17 columns")
  print("Random Forest CV Accuracy:",
        round(rf_grid.best_score_ * 100, 2), "%")
  print("Random Forest Test Accuracy:",
        round(rf_tuned_accuracy * 100, 2), "%")

  print("Gradient Boosting CV Accuracy:",
        round(gb_grid.best_score_ * 100, 2), "%")
  print("Gradient Boosting Test Accuracy:",
        round(gb_tuned_accuracy * 100, 2), "%")

  print("Final Model:", final_model_name)
  print("New Data Prediction:", prediction[0])
  print("Model saved successfully.")

▼ ... =====
TRAFFIC CONGESTION PREDICTION - FINAL SUMMARY
=====
Dataset: 2000 records, 17 columns
Random Forest CV Accuracy: 91.38 %
Random Forest Test Accuracy: 88.75 %
Gradient Boosting CV Accuracy: 95.5 %
Gradient Boosting Test Accuracy: 96.25 %
Final Model: Gradient Boosting
New Data Prediction: High
Model saved successfully.
```



Conclusion

- Successfully predicted traffic congestion as Low, Medium, and High.
- Compared Random Forest and Gradient Boosting models.
- After tuning, Gradient Boosting achieved 96.25% accuracy.
- EDA and preprocessing confirmed good data quality with no missing or duplicate values.
- The model can support data-driven traffic monitoring and congestion prediction.



Our goal is to turn traffic data into actionable predictions for smarter and more efficient transportation.



Thank You

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